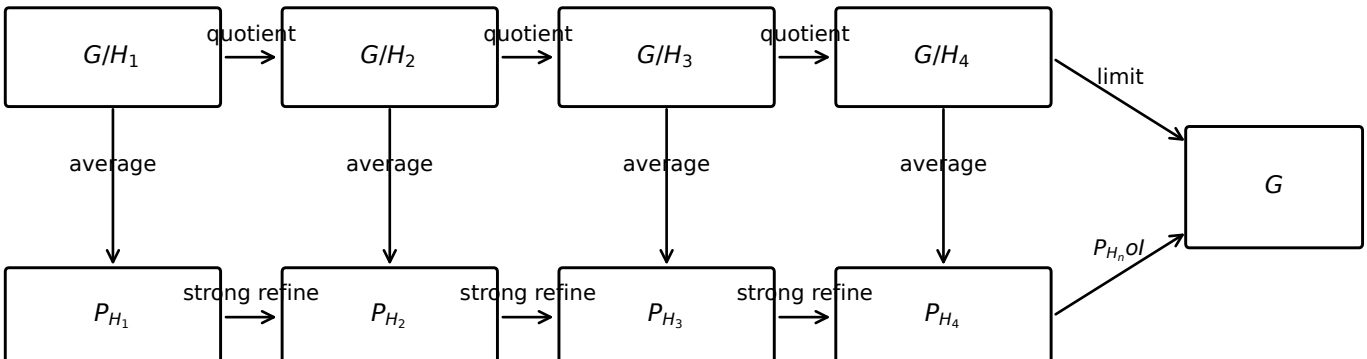Profinite symmetry is reconstructed from finite quotient resolutions